

Elie Jerdak

(Software Engineer & Data Analyst)

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Summary

Computer Science graduate and backend-focused Full-Stack Developer with experience building scalable web applications and APIs using ASP.NET Core and Node.js. Strong understanding of backend architecture, authentication systems (JWT, RBAC), and OOP principles, with additional experience in Python for data-driven and machine learning projects. Familiar with Django and Java, and capable of applying algorithmic problem-solving and AI concepts to academic projects. Comfortable working across the full stack with a focus on backend system design and performance.

Education

- **Bachelor of Science in Computer Science**
Université Saint Esprit Kaslik (USEK) — Graduated with Honors, 2026

Experience

- **Software Engineering Intern | MT2**
June 2025 – July 2025
 - Designed and implemented a full-stack Customer Management System (CMS) for gym facilities using ASP.NET Core and Entity Framework Core, focusing on scalable backend architecture and modular service design.
 - Built a secure authentication system combining JWT and HTTP-only cookies to support API security and session-based MVC access control.
 - Integrated Stripe API for automated payment processing and subscription handling, including backend webhook handling for payment confirmation.
 - Developed a QR-based check-in system for members, implementing secure token generation and validation for controlled access workflows.
 - Deployed the Fitness Club Management System on Render using Docker-based deployment architecture: <https://fitnessclub-cms.onrender.com/>

Projects

- **Car Price Predictor & SoFIFA Data Analysis | Python, Django, Machine Learning**
 - Built a machine learning regression pipeline in Python to predict car prices, deployed through a Django-based web interface for interactive user input and predictions.
 - Performed data preprocessing and feature engineering on the SoFIFA dataset, applying unsupervised anomaly detection techniques including Local Outlier Factor (LOF), k-Nearest Neighbors (kNN) distance-based detection, and PCA-based reconstruction error to identify and handle abnormal data points.
 - Applied SHAP (Shapley Additive Explanations) to interpret model predictions and analyze feature influence on output behavior.
 - Deployed the application publicly on Render: <https://used-car-price-predictor-mr54.onrender.com/>

- **Course Management System | Node.js, Express, EJS**
 - Developed a server-rendered Course Management System using Node.js, Express, and EJS following the MVC architecture, implementing full CRUD functionality for courses, users, and enrollments.
 - Designed a modular backend structure separating concerns across authentication, course management, and administrative operations to improve maintainability and scalability.
 - Implemented server-side rendering to dynamically generate views from backend state without reliance on frontend frameworks.
 - Deployed the application on Render for public access and testing: <https://course-management-1g9y.onrender.com/>
- **Connect-4 AI Engine | Minimax, Alpha-Beta Pruning**
 - Implemented an AI-powered Connect-4 game using the Minimax algorithm optimized with Alpha-Beta pruning for efficient decision-making.
 - Built a simulation system for AI vs AI gameplay to evaluate strategy performance and optimize move selection logic.
 - Developed a visual evaluation system to demonstrate AI decision paths and game state analysis.
 - Deployed the project publicly using GitHub Pages: <https://elie-jerdak.github.io/connect4-ai/>

Skills

- Languages: C#, Java, C++, JavaScript (Node.js), Python, SQL, HTML, CSS
- Backend & Frameworks: ASP.NET Core MVC, Node.js, Express, FastAPI
- Databases: SQL Server, MySQL, MongoDB
- Machine Learning & Data: Scikit-learn, Feature Engineering, SHAP, Unsupervised Anomaly Detection
- Backend & System Design: REST APIs, JWT Authentication, MVC Architecture, OOP Principles
- Tools & Platforms: Git, Stripe API, Power BI
- CS Fundamentals: Data Structures & Algorithms

Languages

- Arabic: Native
- English: Fluent (Professional Working Proficiency)
- French: Working Proficiency

Note:

- **The deployed applications may take a few seconds to start because they are hosted on Render's free tier.**
- **Please use the following credentials should you like to test them:**
 - **Fitness-Club:**
 - **Email:** jardakelie@gmail.com
 - **Password:** Password_123
 - **Course Management System**
 - **Admin :**
 - **Email:** elie@email.com
 - **Password:** Password_123
 - **Instructor:**

- **Email:** philippe@email.com
- **Password:** Password_123
- **Student:**
 - **Email:** maroun@email.com
 - **Password:** Password_123